

0.1 Overview

The NA62 experiment is a fixed target experiment located in the north area of CERN with a high energy particle beam provided by the Super Proton Synchrotron. The main goal of the experiment is to measure the branching fraction of the ultra-rare charged kaon decay $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ which is of the order of 10^{-10} to a 15% precision. Alongside there is a expansive physics program of precision and rare decay measurements as well as searches for forbidden processes and exotic particles.

The experiment employs a decay-in-flight technique to obtain a large number of charged kaon decays, in the order of 10^{13} within the short time frame of a few years. The experimental setup was designed to fulfil all the requirements for the challenging measurement of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$, which involves the detection of the K^+ upstream and the detection of the product π^+ , with a missing energy-momentum carried away by the undetected neutrino-antineutrino pair. The measurement of the missing mass is used for kinematic rejection of the main background K^+ decays; $K^+ \rightarrow \pi^+ \pi^0$, $K^+ \rightarrow \pi^+ \pi^- + \pi^0$ and $K^+ \rightarrow \mu^+ \nu_\mu$ by selecting two regions shown in figure 1. The missing mass calculation requires precise timing and position measurement of the K^+ and π^+ candidates, provided by beam (section 0.3.2) and secondary (section 0.3.4) particle spectrometers. The measured 3-momenta of the candidates allows for the reconstruction of the 4-momenta of the Kaon P_{K^+} and the pion P_{π^+} which is used in equation 0.1

$$m_{miss}^2 = (P_{K^+} - P_{\pi^+})^2 \quad (0.1)$$

The rest of this chapter will cover how NA62 fulfils the above mentioned requirements. Section 0.2 will describe the NA62 beam-line, upstream of the NA62 detector. The NA62 detector comprises a number of sub-detectors, which will be described in Section ??, with the trigger and data acquisition system (TDAQ) following in Section 0.4

0.2 The NA62 Beam Line

The experiment at CERN uses a hadron beam extracted from the SPS with a nominal momentum of $400 \text{ GeV}/c$, directed onto a 400 mm long, 2 mm diameter beryllium rod that forms the fixed target (T10) located at 0 m in figure 3. The beam derived

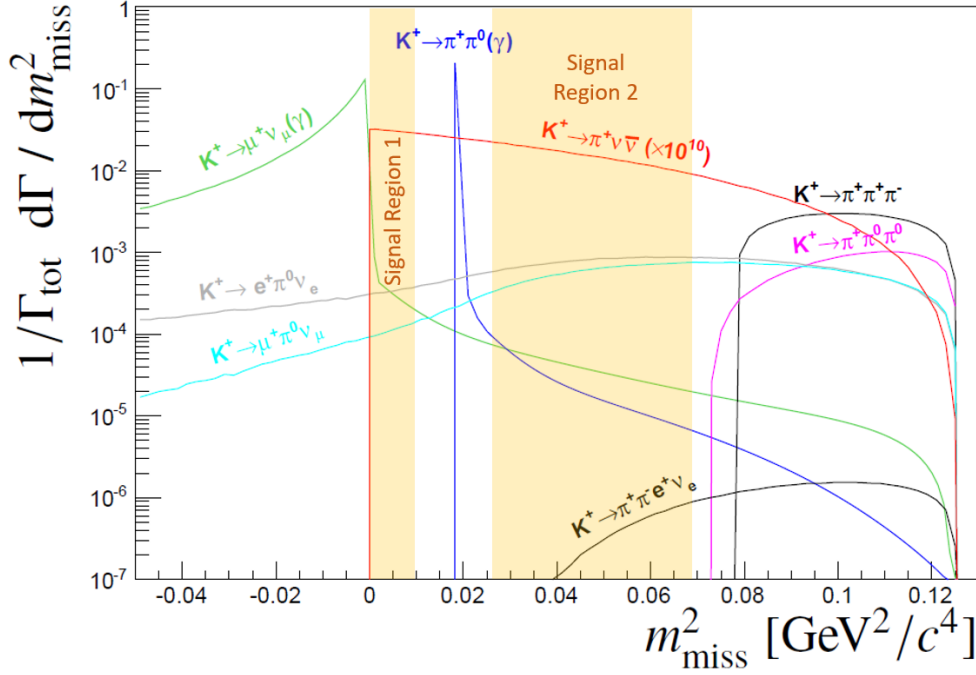


Figure 1: Expected squared missing mass distributions of main kaon decays and $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ (scaled by 10^{10} for for visibility). The solid coloured areas show the signal regions used to reject the dominant kaon decays kinematically.

from the T10 target, referred to as K12, is a high-intensity unseparated hadron beam with a nominal momentum of $75 \text{ GeV}/c$. The beam momenta was chosen to maximise the fraction of kaons with respect to the other beam components at the entrance to the detector.

The elements of the K12 beam-line are shown in Figure 3 and comprise a number of components. First are three quadrupole magnets (Q1, Q2 and Q3), used to focus the beam with a large solid angle acceptance of $\pm 2.7 \text{ mrad}$ horizontally and $\pm 1.5 \text{ mrad}$ vertically at a $75 \text{ GeV}/c$ central momentum. Next follows an achromat (A1), which selects a beam of a $75 \text{ GeV}/c$ momentum with a 1% rms momentum. The A1 achromat comprises four dipole magnets that vertically deflect the beam, the first two which deflect the beam 110 mm downwards, while the final two deflect the beam upwards towards the original axis. Between these pairs of magnets, there are a pair of water-cooled beam-dump units (TAX1 and TAX2) with a set of holes to select the required $75 \text{ GeV}/c$ momentum beam, while absorbing unwanted secondary beam particles and any remaining primary protons. TAX1&2 can be closed to act as a secondary target for use in dump mode or for safety when access is

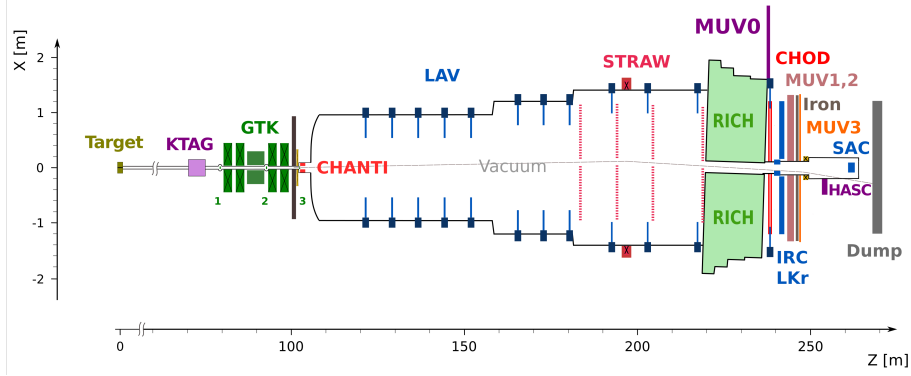


Figure 2: X-Z diagram of NA62 detector setup for Run 1

required downstream. At the focus between the TAX1&2, a radiator consisting of tungsten plates with a thickness of up to 5 mm is introduced into the path of beam. The radiator is optimised to reduce positron energy while minimising the loss of energy to hadrons. A triplet of quadrupoles (Q4, Q5, Q6) follows, which refocuses the beam in the vertical dimension and produces a parallel beam in the horizontal dimension. The space between the quadrupoles are occupied by a pair of collimators (C1, C2), which redefines the horizontal and vertical acceptance for the beam. The collimator after the Q6 quadrupole (C3) is employed to absorb positrons that had their momenta degraded by the radiator and again redefines the acceptance in the vertical dimension.

The beam next passes through a 40 mm field-free hole in a set of iron plates, placed between three 2 m dipole magnets (B3). The vertical magnetic field in the iron plates is used to deflect muons of both signs away from the beam. Any deviation in the beam due to stray fields in the beam hole are canceled out by a pair of steering dipoles (TRIM2 and TRIM3). Two quadrupoles (Q7, Q8) follow and ensure that the beam is parallel before the KTAG which employs a differential Cherenkov counter (CEDAR), described in Section 0.3.1, which requires the beam to be parallel. Before the KTAG, a pair of collimators (C4 and C5) are used to absorb particles in the tails of the beam.

For beam monitoring, two pairs of filament scintillator counters (FISC 1, 3 and FISC 2, 4) are installed at either end of the KTAG. Used in coincidence, the beam divergence can be measured and used to tune the divergence to zero.

Following the KTAG are two quadrupoles (Q9 and Q10), used to focus the beam before the beam tracking system (GTk). The GTk comprises four stations made up

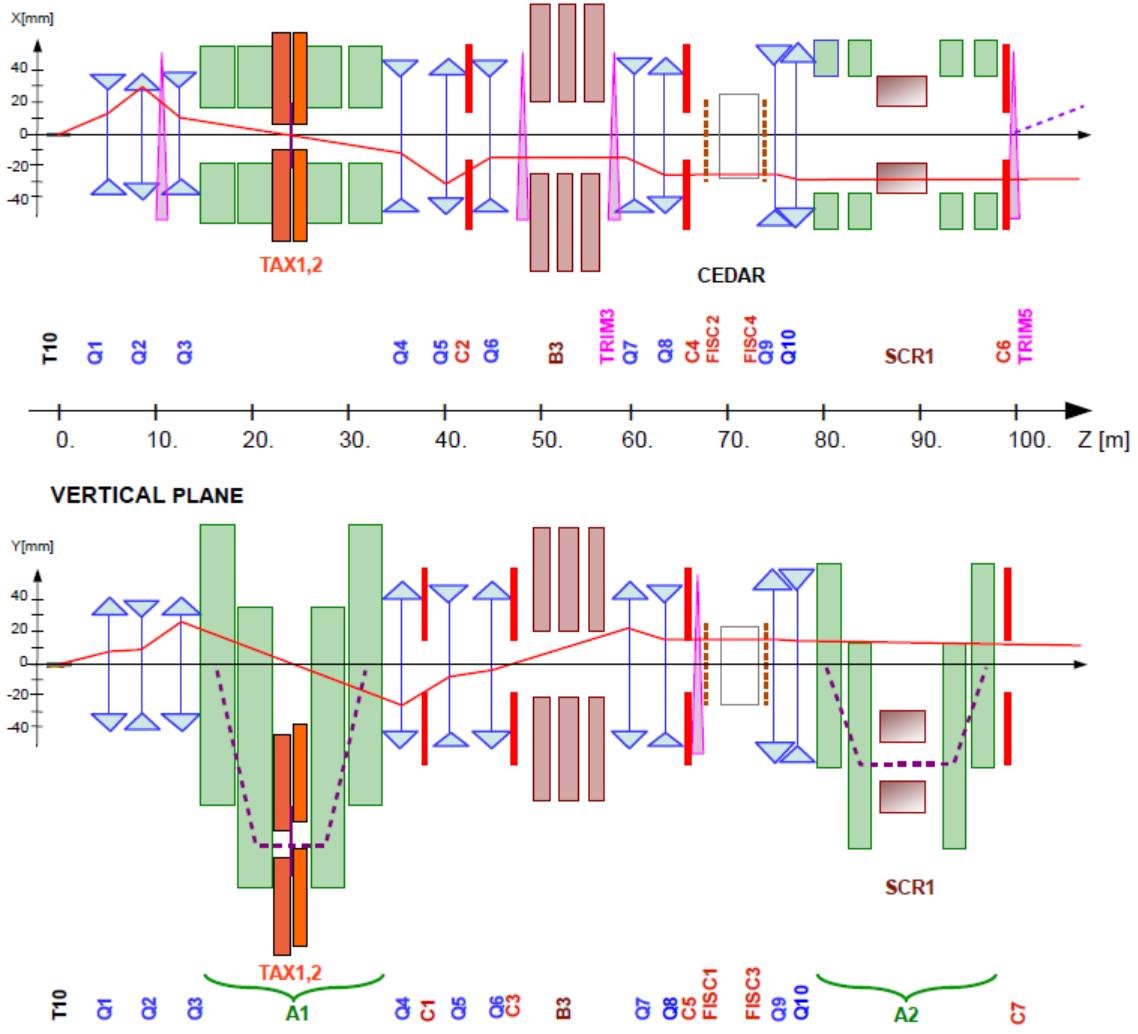


Figure 3: Schematic of K12 beamline from T10 target at 0m to the entrance of the NA62 decay region at 102.4m[1].

of silicon pixel detectors described in Section 0.3.2 installed in the beam vacuum. The GTK stations are installed so that the space between GTK1 and GTK3 is occupied by a second achromat (A2), which comprises four dipole magnets that deflect the beam 60 mm vertically, allowing for the beam particle momenta to be measured. The return yokes of the third and fourth magnets and a magnetised iron collimator (SCR1), defocus beam muons which leave the beam between the second and third magnets. A pair of cleaning collimators (C6 and C7) used to absorb beam particles outside of the acceptance are located before the final GTK station (GTK3), which is at the entrance to the decay region ($Z = 102.4\text{ m}$). A steering magnet (TRIM5) is then used to deflect the beam in the horizontal dimension by an angle of $+1.2\text{ mrad}$.

At $Z = 102.4\text{ m}$ begins a large 117 m vacuum pipe (Bluetube), evacuated to a

pressure of 10^{-6} mbar by several cryo pumps along its length. The Bluetube is made up of 19 cylindrical sections ranging in diameter shown in Figure 2, from 1.92 m in the first section after GTK3, to 2.4 m in the middle section and then 2.8 m in the region containing the STRAW spectrometer. The first section is defined as the decay region between $Z = 105\text{ m}$ and $Z = 170\text{ m}$. Further downstream, the Bluetube hosts a number of detectors: Eleven LAV stations used in the Photon detection system described in Section 0.3.6 and four STRAW spectrometer chambers coupled with a large aperture dipole magnet (MNP33), providing kinematic measurements of the charged products of kaons decaying in the decay region, described in Section 0.3.4. The MNP33 magnet provides a $270\text{ MeV}/c$ momentum kick in the horizontal dimension, deflecting the $75\text{ GeV}/c$ beam by -3.6 mrad , so that combined with the TRIM5 deflection, the beam passes through the central aperture of the Liquid Krypton calorimeter (LKr). The path of the beam is shown in Figure 4, showing the deflection in the horizontal dimension. After the final STRAW spectrometer chamber (STRAW4), the Bluetube is closed off with a thin aluminum window (except for the hole for the beam to pass through), separating the vacuum from the 17 m long, neon-filled Ring Imaging Cherenkov counter (RICH), designed to distinguish between pions and muons produced in kaon decays and is part of the Particle Identification system (PID) and is described in Section 0.3.5. The beam is transported through the remaining downstream detectors in vacuum and is deflected -13.2 mrad in the negative direction by a dipole magnet (BEND) to sweep the beam away from the Small Angle Calorimeter (SAC) acceptance, located within the vacuum beam pipe. Finally, the beam is absorbed in a beam dump.

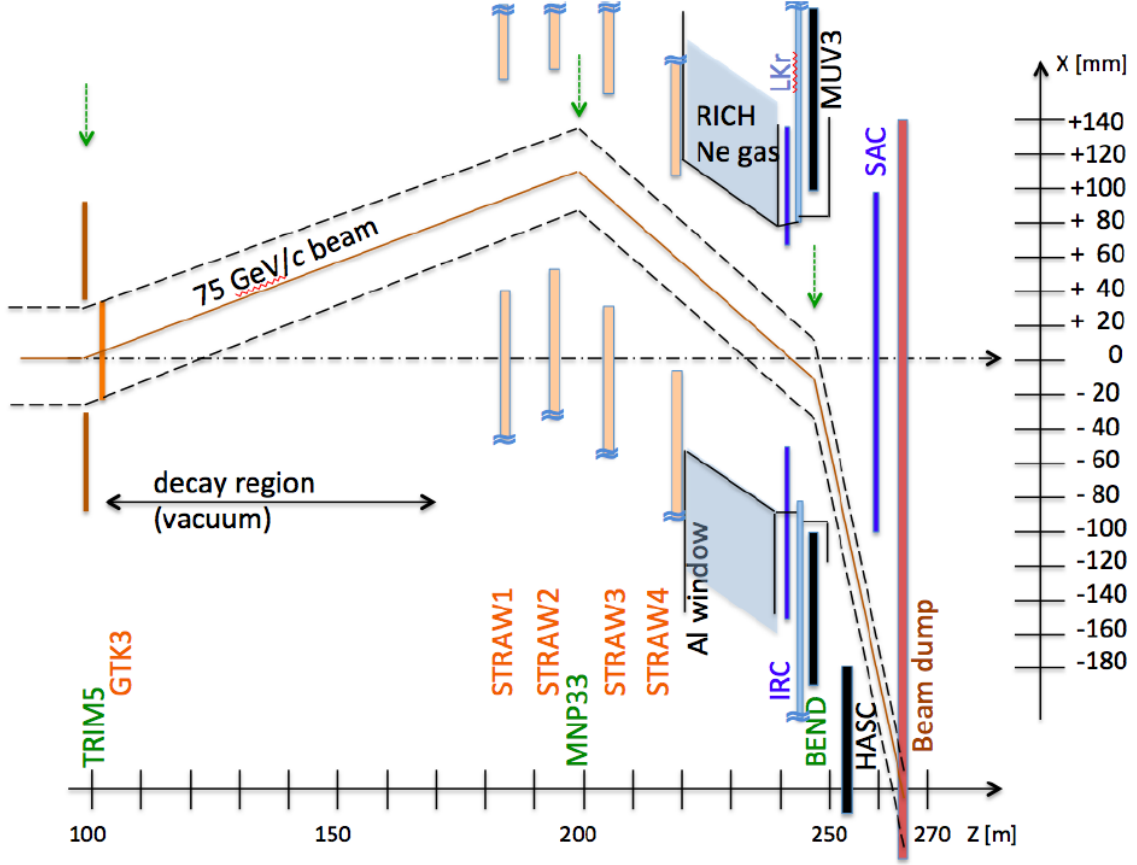


Figure 4: Schematic of K12 beamline after GTK3 [1].

0.3 NA62 Detectors

0.3.1 KTAG

As the NA62 beam is unseparated and kaons only account for 6% of the beam components, the kaons must be identified. Kaon identification is provided by the KTAG and located at $Z = 69.4\text{ m}$. It comprises a CEDAR vessel and a purpose-built photon detection system. Until the NA62 2023 run, the CEDAR vessel used was a CEDAR-W filled with a Nitrogen radiator. CEDARs were developed at CERN in the 1970s [2] for secondary beam particle detection at the super proton synchrotron (SPS). The CEDAR is a differential Cherenkov Counter and exploits the dependence of the beam particle's mass on the properties of the produced light, defined in equation 0.2 where $\beta = \frac{p}{\gamma m}$.

$$\cos(\theta) = \frac{1}{\beta n} \quad (0.2)$$

In a beam of set momentum and a radiator with a constant refractive index, the angle of Cherenkov light is only affected by the mass of the beam particle. The pressure is set at 1.7 bar of Nitrogen in the CEDAR-W so that the light from the kaon is focused by a set of lenses and mirrors onto arrays of photomultiplier tubes (PMTs). The light from the other beam components does not reach the PMT arrays. A more detailed description of CEDARs can be found in Section ??.

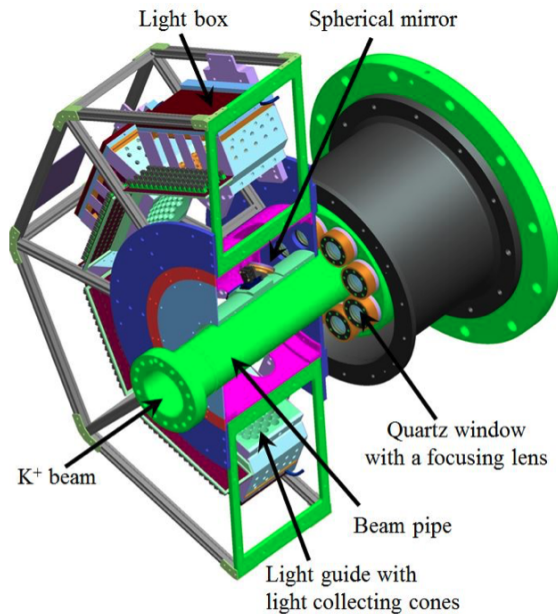


Figure 5: Model of the upstream part of the CEDAR and the KTAG photon-detection system [1].

In the original CERN design, the Cherenkov light exiting each CEDAR exit window was detected by an ET-9820QB photomultiplier tube (PMT) placed directly on the exit window, with a particle identified if there was a coincidence of at least six PMTs. However, these PMTs would not fulfil the timing (≤ 100 ps) and rate requirements (45 MHz of kaons) at NA62.

For NA62, a purpose-built photon-detection system shown in Figure 5 was designed, constructed and mounted to the upstream end of the CEDAR Vessel. The photon light from the CEDAR exit windows are reflected radially 90 degrees by eight spherical mirrors onto eight PMT plane, referred to as sectors. Each PMT plane is equipped with an array of 48 single-anode Hamamatsu PMTs of two types: 32 R9880-110 and 16 R7400. The signals from the PMT arrays are clustered into kaon candidates within a nominal 2 ns time window. A kaon was identified when there are at least five sectors in coincidence (5-fold coincidence). The clustering results

with on average 18 photo-electrons per kaon candidate. The PMTs can achieve time a resolution of 300 ps resulting in a reconstructed kaon time resolution of 70 ps and a kaon identification efficiency of 95% was achieved with the KTAG and CEDAR-W [3].

In 2023, a new CEDAR with a Hydrogen radiator (CEDAR-H) was installed in the NA62 beam-line after design at the University of Birmingham and testing at CERN. The new CEDAR uses the same basic principles as previous CEDARs but uses Hydrogen as a radiator gas, which required the design of new optical elements in the CEDAR and the photon detection system. A more detailed description of the CEDAR and its optical elements, along with the design, testing and installation of the new CEDAR-H can be found in Section ??.

0.3.2 GigaTracker (GTK)

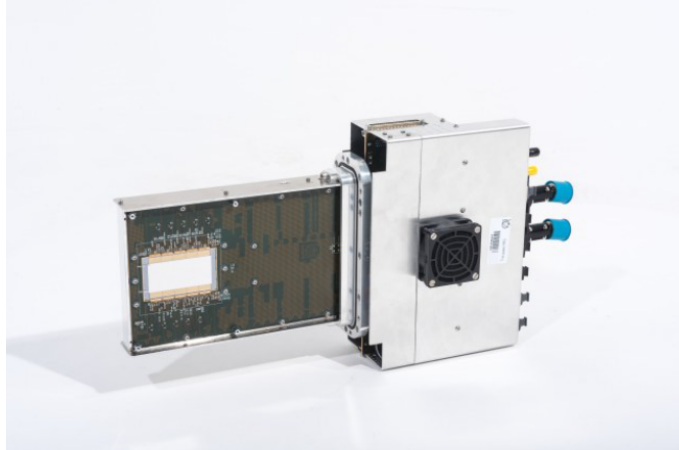


Figure 6: Image of GTk station [1].

The precise measurement of the beam momentum, time, and position is provided by the GigaTracker (GTK), a beam spectrometer comprising hybrid silicon trackers placed within the vacuum pipe along with an achromat upstream of the decay volume. As particles of differing momenta are deflected by different amounts in a magnetic field, the momentum of a particle can be measured by comparing the positions of the particles in each station. The resolution of the momentum measurement is,

$$\frac{\Delta p}{p} = 0.2 \% \quad (0.3)$$

with a direction at the exit of the achromat is $16 \mu\text{rad}$ and a hit time resolution of 200 ps . During run one data-taking period (2016-2018), three stations were employed and are shown in Figure 7. Another station was added (GTK0) for run2 data taking (2021-present) and placed in front of the first GTK station, increasing the tracking efficiency and reducing the backgrounds for the main decay mode. The layout of the GTK is shown in Figure ??

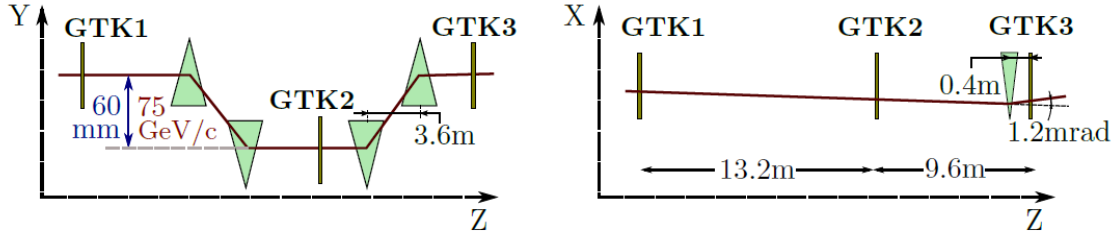


Figure 7: Schematic of GTK stations within the acromat [1] for run 1.

Each station consists of 18000 pixels of size $300 \times 300 \mu\text{m}^2$ in a matrix of 200×90 , covering a total area of $62.8 \times 27 \text{ mm}^2$ and is read out by application-specific integrated circuits (ASIC). The tracker was placed within the beam, so the material has to be minimised to reduce the number of interactions with the beam and improve the angular resolution. The material budget for each station was set at 0.5 %, corresponding to $500 \mu\text{m}$ of Silicon.

0.3.3 Charged Anti-coincidence detector (CHANTI)

The CHANTI is located after the GTK and its primary goal was to provide rejection of particles produced from inelastic collisions of beam particles with the final GTK station. Vetoing these particles is imperative as they would enter the decay volume and result in background for $K^+ \rightarrow \pi^+ \nu \bar{\nu}$. The detector was also used to detect the muon halo close to the beam. The CHANTI was a set of six square hodoscope stations comprising of two planes, one paced vertically and the other horizontally, with a fully constructed station shown in Figure 8. The planes were comprised of polystyrene based scintillator bars with central fast wavelength-shifting (WLS) fibre which were then connected to silicon photomultipliers (SiPMs).

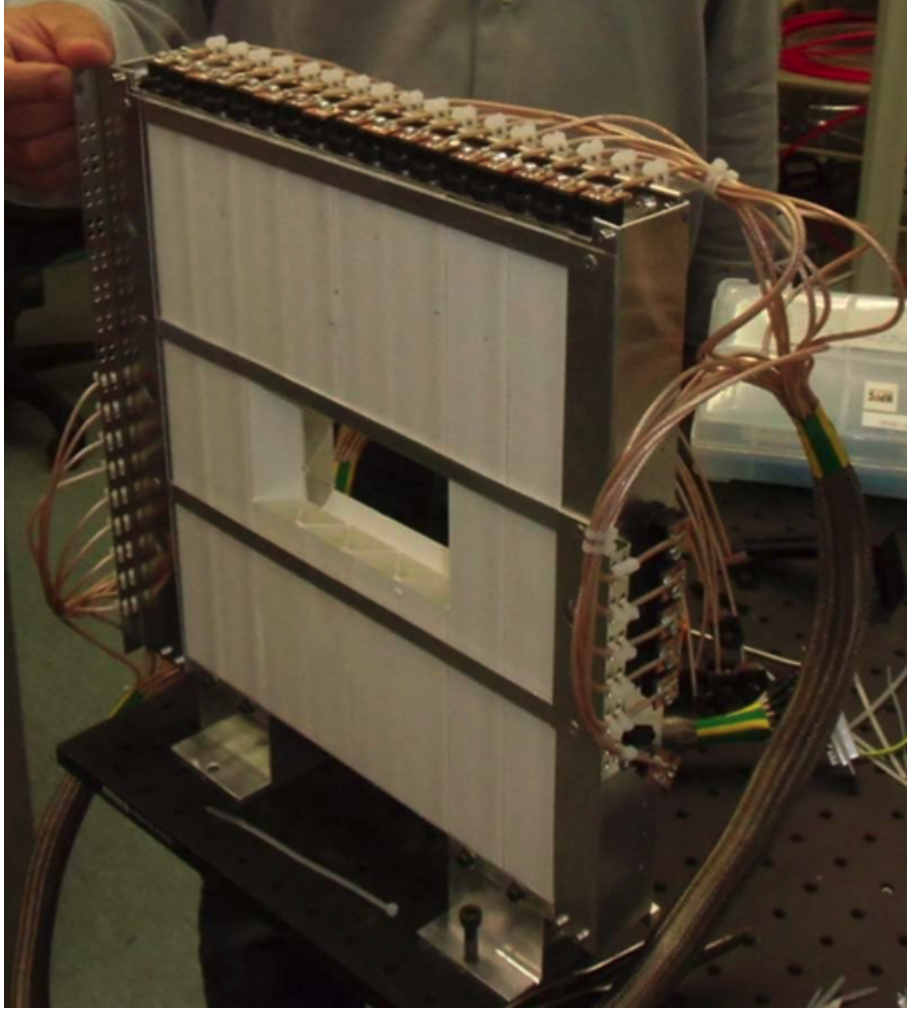


Figure 8: Image of a single CHANTI station [1].

0.3.4 Straw Spectrometer (Straw)

The STRAW Spectrometer provides kinematic measurements of charged decay products produced in K^+ decays within the decay volume. It comprises four chambers and a large aperture dipole magnet (MNP33), providing a momentum resolution of

$$\frac{\sigma(p)}{p} = 0.30\% \oplus 0.005\% \quad (0.4)$$

where p is the track momentum in GeV/c . The chambers are downstream from the decay region, with the first chamber located at $Z = 183\text{ m}$ and the second 10 m further downstream. The MNP33 magnet is situated downstream of chamber 2 at $Z = 197.6\text{ m}$ and provides an integrated field of 0.9 Tm . Chambers 3 and 4 are located downstream of the MNP33, $Z = 204\text{ m}$ and 219 m respectively.

Each chamber comprises four views (X, Y, V, and U) rotated at 0 deg, 90 deg,

+45 deg, and -45 deg, each consisting of four layers of straws shown in Figures 9 and 10. The views have a gap near the center, of around 12 cm with no straws, such that when they are combined to form a chamber, an octagonal hole with an apothem of 6 cm is produced for the beam to pass through. As shown in Figure 4, the beam hole for each chamber is not centred to the angle of the beam before and after the MNP33 magnet, with the beam hole parameters defined in Table 1.

Chamber	Beam Hole Size (cm)	Beam Hole Offset (X, Y, Z)
1	6	101.2 mm, 0 mm, 183 m
2	6	114.4 mm, 0 mm, 193 m
MNP33	-	-, -, 198 m
3	6	92.4 mm, 0 mm, 204 m
4	6	52.8 mm, 0 mm, 218 m

Table 1: Straw spectrometer chamber beam-hole positions

The straws are 2.1 m long drift tubes that are 9.82 mm in diameter, made from 36 μ m thick polyethylene terephthalate (PET), coated on the inside with 50 nm of copper and 20 nm of gold. The Straws are filled with a mixture of 70% Argon and 30% CO_2 at atmospheric pressure, and have a central gold-plated tungsten anode wire. The straws act like proportional counters, when a charged particle passes through the gas, it is ionised, creating electron-ion pairs that drift to the opposite electrodes inducing a charge on the anode wire. When the electron approaches the anode wire, the electric field strength increases dramatically, producing Townsend avalanches. This provides a large multiplication of the signal, producing a charge pulse, which can be read out.

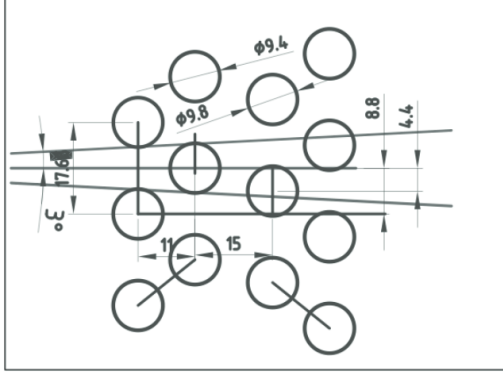


Figure 9: Diagram of straw placement in each view [4]

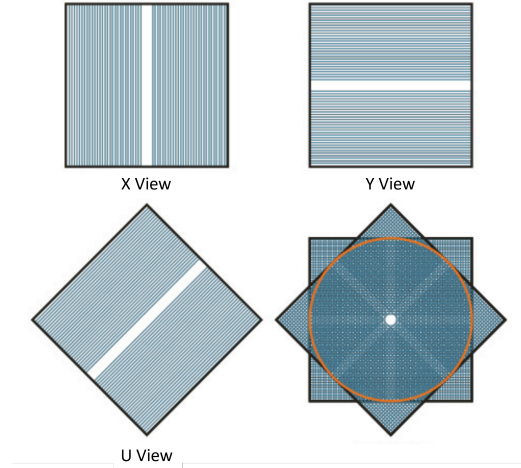


Figure 10: Diagram showing placement of views in a straw chamber. Edited from [4]

0.3.5 Ring Imaging Cherenkov counter (RICH)

The Ring Imaging Cherenkov counter (RICH) was designed to separate secondary pion and muons within a momentum range of 15 - 35 GeV/c . The ability to separate the secondary particles is required as the largest background to $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ comes from the $K^+ \rightarrow \mu^+ \nu_\mu$, which has a BR 10 orders of magnitude greater. The majority of the suppression comes from kinematic selections and the response difference between pions and muons in the calorimeters, however, another factor of 100 is required in the momentum range 15 - 35 GeV/c was required, which the RICH satisfied.

The RICH is a 17.5 m long cylindrical vessel comprising four steel sections with diameters gradually decreasing from 4.2 m at the upstream section to accommodate the PMT flanges to 3.2 m in the downstream end. The radiator vessel is separated from the vacuum decay tube by an aluminum window of 2 mm thickness and radius of 1.1 m at the upstream end, allowing minimal material in the path of particles entering the detector. A lightweight aluminum pipe connected to the vacuum tube passes through the vessel, allowing the beam to pass through in a vacuum and continues through the 4 mm thick, 1.4 m radius aluminum exit window. The vessel contains Neon gas at a pressure of 900 mbar with a refractive index of $n = 1 + 62.8 \times 10^{-6}$ at

a wavelength of $300\text{ }\mu\text{m}$, corresponding to a Cherenkov threshold for charged pion of $12.5\text{GeV}/c$.

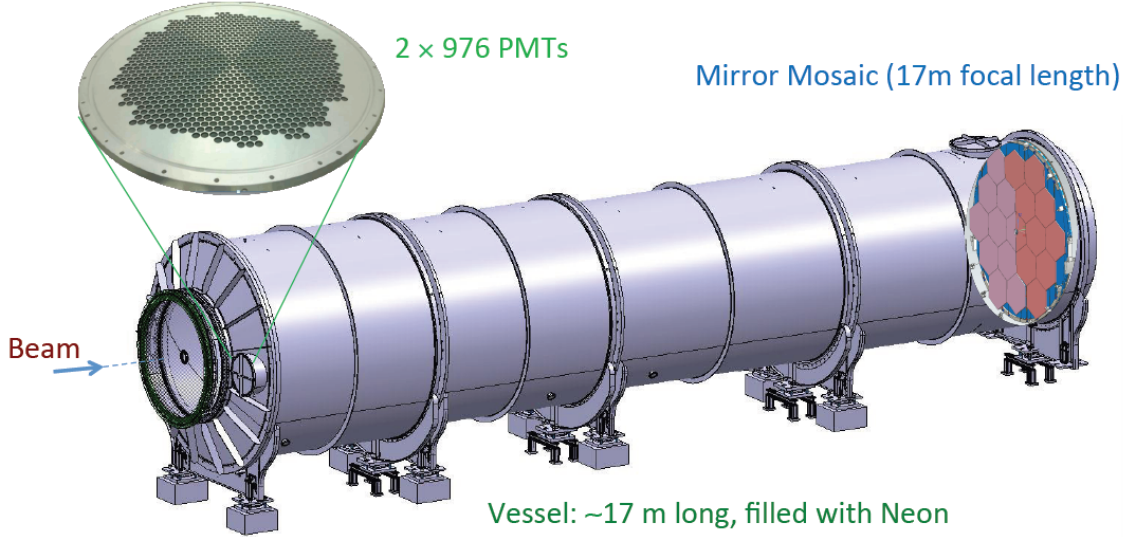


Figure 11: Schematic of RICH detector, showing the beam entering the upstream end on the left. A zoomed-in image of one of the two PMT flanges shows the layout of the PMT arrays. The mirrors are also shown at the downstream end of the vessel [5]

A mosaic of 20 spherical mirrors (18 hexagonal and 2 semi-hexagonal) with a focal length of 17 m are located at the downstream end of the vessel and is shown in Figure 11. This mirror system is used to reflect and focus the Cherenkov light toward two PMT arrays located at the upstream end of the vessel. The two PMT arrays are located on either side of the entrance window (see Figure 11), each consist of 976 Hamamatsu R7400-U03 PMTs. Each PMT has a 8 mm active area and are packed into a hexagonal lattice, each with a Winston cone directing light onto the active area.

The particles are identified (selected or rejected) using a cut on the particle's mass, calculated from the particle velocity using the reconstructed Cherenkov ring radius shown in Figure 12 and the momentum of the particle measured by the spectrometer. In Figure 12, it is clear that separating pions and muons within the desired momentum range is possible.

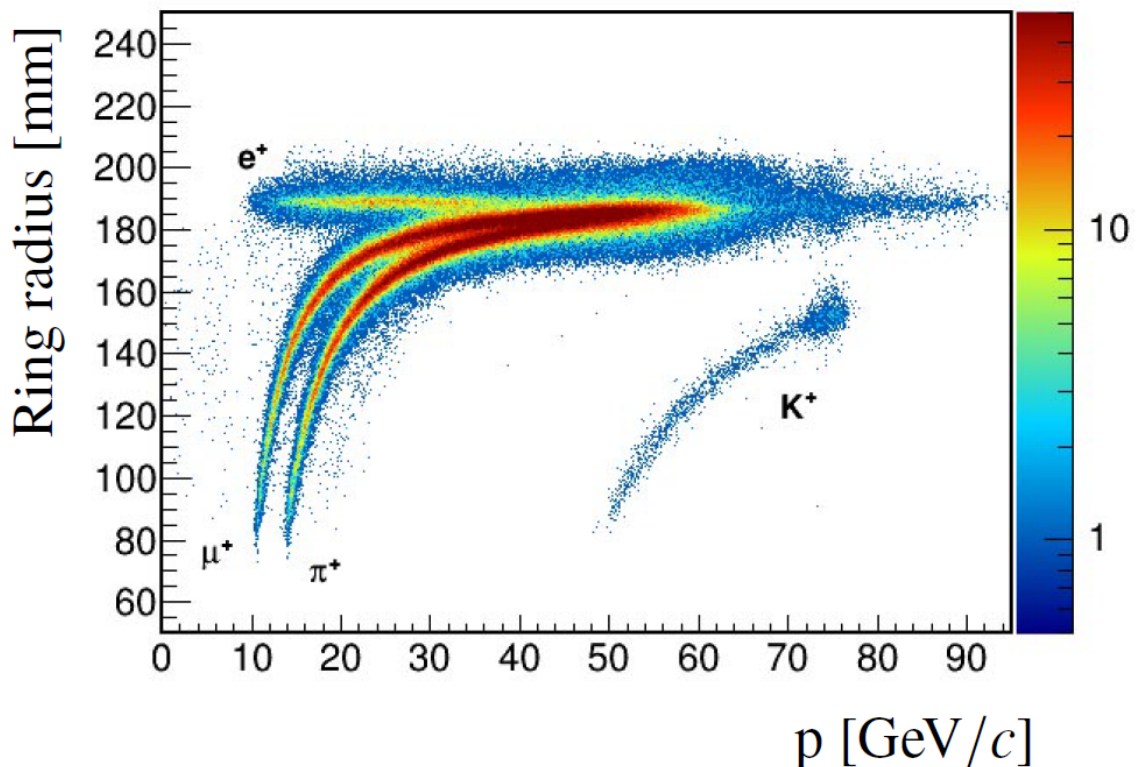


Figure 12: Cherenkov ring radius as a function of the particle momentum [5]

0.3.6 Photon Detection System

The photon detection system provided a rejection power of 10^8 for the main decay background $K^+ \rightarrow \pi^+\pi^0$. It was made up of four calorimeters used to detect and reject background decays with photons in the final state, with coverage from 0 to 50 mrad. Photons with small production angles between 0 and 1 mrad were covered by the intermediate-ring calorimeter (IRC) and small-angle calorimeter (SAC).

Photons with production angles between 1 and 8.5 mrad were covered by the NA48 liquid krypton calorimeter (LKr). A quasi-homogeneous calorimeter which consisted of a large cryostat containing 9000 litres of liquid krypton, which extends from the beam-pipe to a radius of 128 cm and had a depth of 127 cm. This corresponded to a radiation length of $27 X_0$, designed to allow for the full capture of electromagnetic showers produced. Thin beryllium-copper ribbons made up the 13248 longitudinal cells, depicted in Figure 13, consisting each of a central anode and cathodes either side. Each anode ribbon was connected at the downstream end

to preamplifiers and the high voltage of 3000V to provide the drift field required for the ionisation.

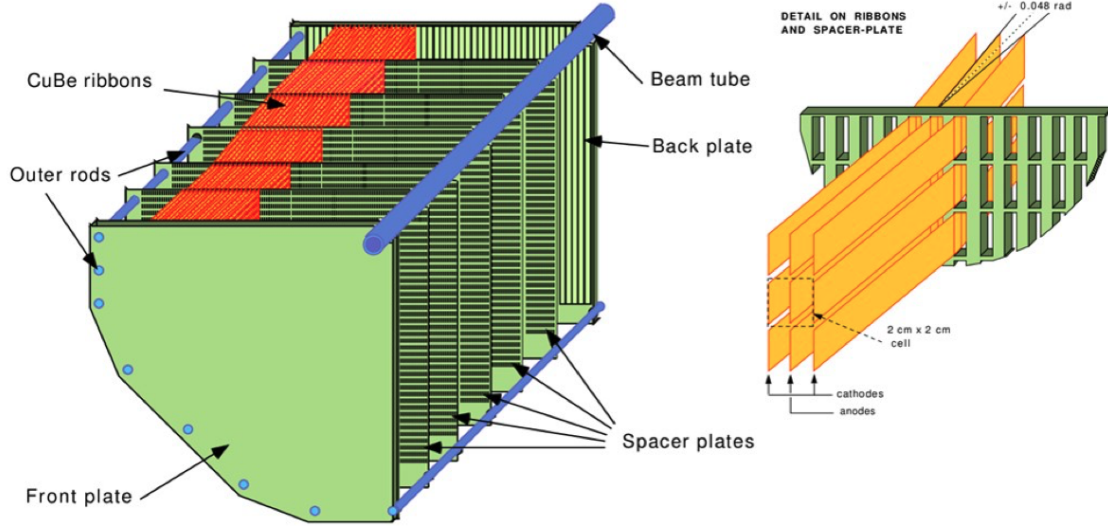


Figure 13: Left, schematic of the structure of a single quadrant of the Lkr. Right, layout of cells within the quadrant. [5]

The Large Angle Veto (LAV) provide coverage from 8.5 to 50 mrad with 12 stations positioned along and beyond the decay volume. 11 of the LAV stations were along the vacuum volume with the final station placed just upstream of the LKr. A LAV station consisted of a barrel of lead glass blocks surrounding the perimeter of the vacuum volume and were recycled from the OPAL experiment [6], with LAV1 picture in Figure 14. The electromagnetic showers produced in the each of the glass blocks resulted in the emission of Cherenkov light, which was then detected by a Hamamatsu R2238 76 mm diameter photomultiplier attached to the back side.



Figure 14: Photo of the installation of LAV1 at NA62 in 2008.

0.3.7 Charged Particle Hodoscopes

The charged particle hodoscopes comprised of pair of scintillating detectors which cover the acceptance downstream of the RICH and upstream of the LKr. The main function of the charged particle hodoscopes was to provide input to the first level of NA62 triggers system. The NA48-CHOD which was used in NA62's predecessor experiment NA48 [7] was located upstream of LAV12, just after the RICH. The NA48-CHOD consisted of two planes of 20 mm thick plastic scintillator slabs, 64 vertical and 64 horizontal which is shown in Figure 15. Each scintillating slab was individually connected to photomultipliers via a light guide. A signal in both planes was required to produce a track time with a resolution of 200 ps. The charged particle rate at the NA48-CHOD was 13 MHz, with the overall rate at 35 MHz mainly resulting from particles produced from interactions with upstream detectors.

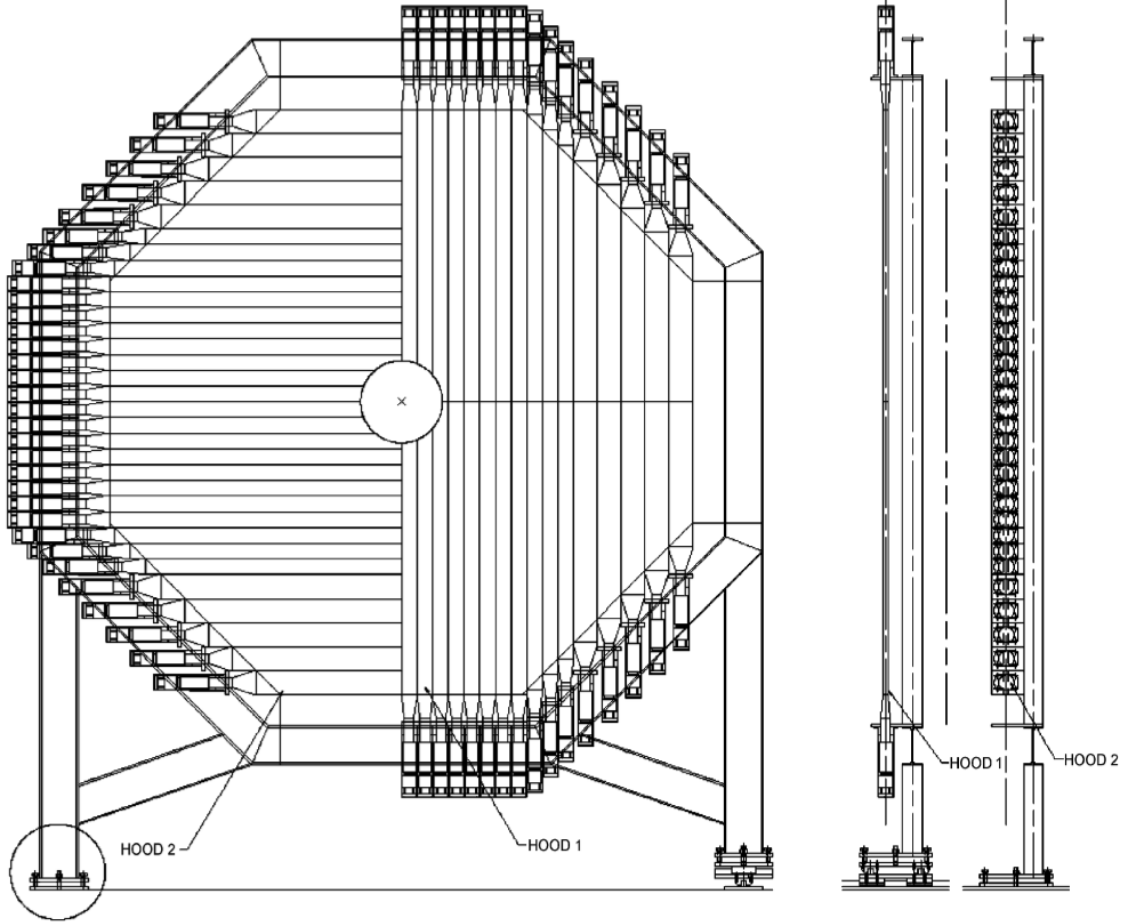


Figure 15: Schematic of the NA48-CHOD [7].

The CHOD was located after the twelfth LAV station, before the LKr and consisted of an array of 152 30 mm thick plastic scintillating tiles that cover an acceptance radius of between 140 and 1070 mm, shown in Figure 16. The scintillating tiles overlap by 1 mm and are supported by a glass fibre foil (G10). The light produced from each tile is collected with a WLS fibre and are bundled into groups of four and are read out by Silicon photomultipliers (SiPM). The overall time resolution of the detector was 1.1 ns. In operation the CHOD would sustain a rate of 45 MHz, with 13 MHz from charged particles.

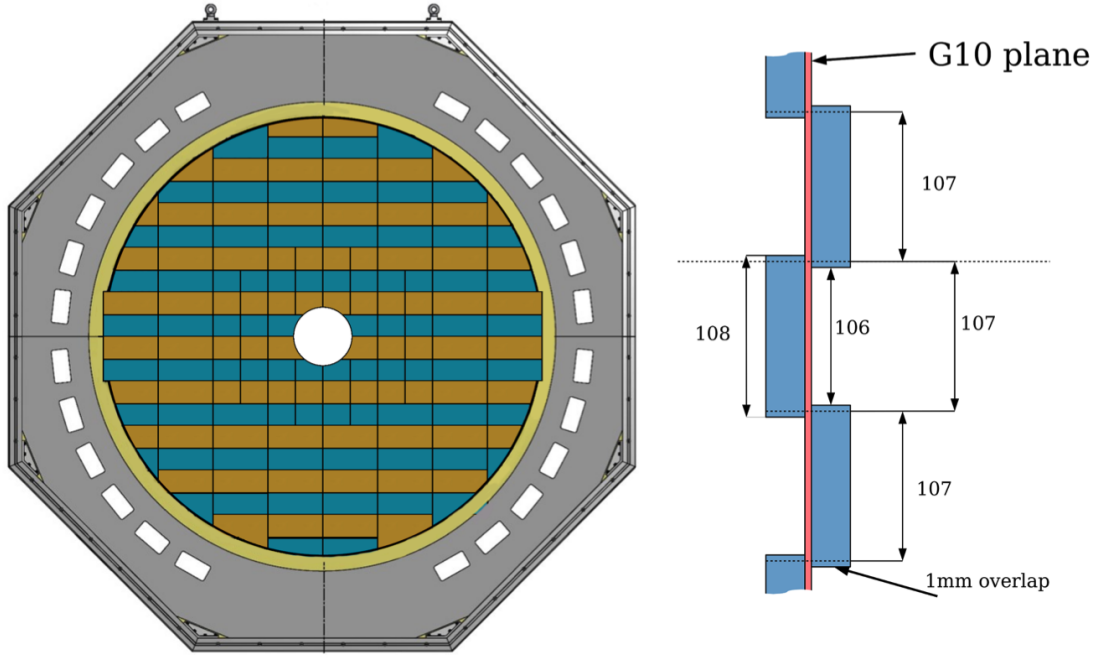


Figure 16: Layout of the NA62 CHOD. Left, the front cross section of scintillating tiles with the support structure. Right, side cross section of the scintillating tiles [8].

0.3.8 Muon Veto System

The Muon Veto System (MUV) was used to discriminate between pions and muons in conjunction with the RICH and LKr. The system comprises of a hadronic sampling calorimeter made from two modules, MUV1 and MUV2 utilising iron as an absorber material. MUV1 was a newly designed module for NA62 and MUV2 was repurposed from the front of the NA48 hadronic calorimeter [7]. MUV1 consisted of 24, 26.8 mm thick iron plates with 23 layers scintillator strips placed of 9 mm thickness with a width of 60 mm, placed in between, shown in the schematic in Figure 17. The scintillating strips alternate between horizontal and vertical, resulting in 12 horizontal and 13 vertical. The total material MUV1 produced a total thickness of $4.1 X_0$ (interaction lengths). The light produced within the strips was collected by WLS fibres and readout by PMTs.

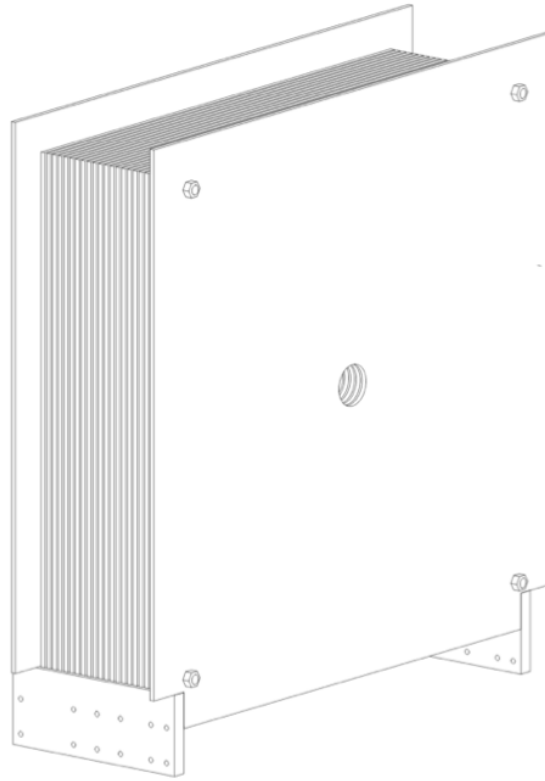


Figure 17: Schematic of the MUV1 module used as part of the Muon Veto System at NA62.

MUV2 was similar in construction with 2×22 strips per plane versus 48 for MUV1, as shown in Figure 18. MUV2 had thinner iron plates at 25 mm thick which resulted in a lower total thickness of $3.7 X_0$.

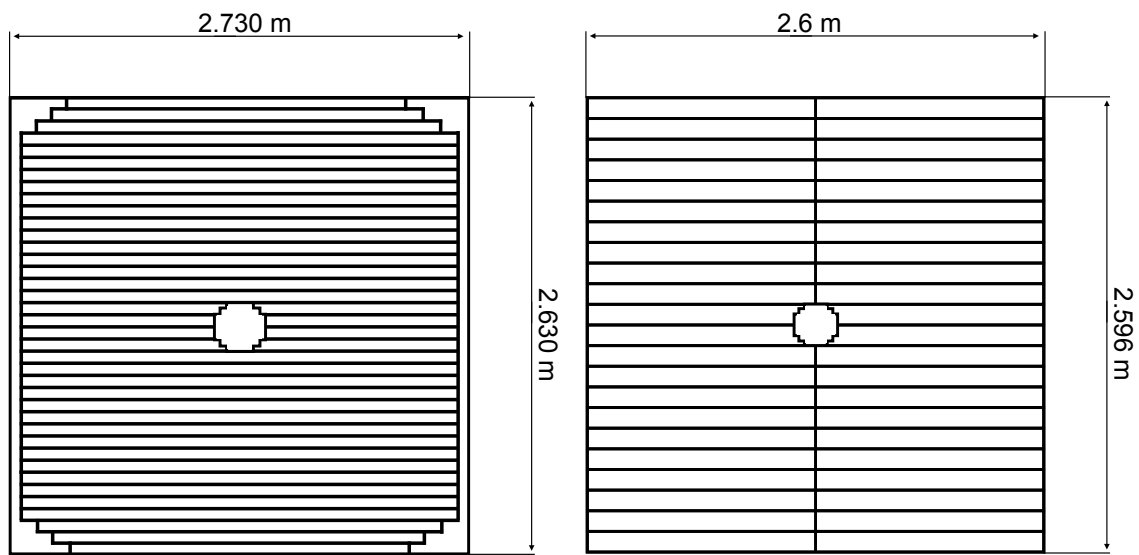


Figure 18: Layout of scintillating tiles for a horizontal layer for: Left, MUV1 and Right, MUV2.

Downstream of the hadronic calorimeter the MUV3 detector was located behind a 80 cm thick iron wall and was used for muon identification. The MUV3 comprises of 140, 59 mm thick scintillating tiles and eight smaller tiles around the beam pipe to account for the high particle rate from the beam. Each tile is readout directly by two PMTs, without any WLS to improve time resolution of the detector to 0.6 ns. The overall muon rate within the detector a nominal beam intensity was 13 MHz. The MUV3 was primarily used as a veto within the L0 trigger system to veto events that include a muon.

0.4 Trigger and Data Acquisition System (TDAQ)

0.5 NA62 Results

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